

REVERSIBLE REACTIONS

Not all chemical reactions proceed to completion. In most reactions two or more substances react to form products which themselves react to give back the original substances. Thus A and B may react to form C and D which react together to reform A and B.



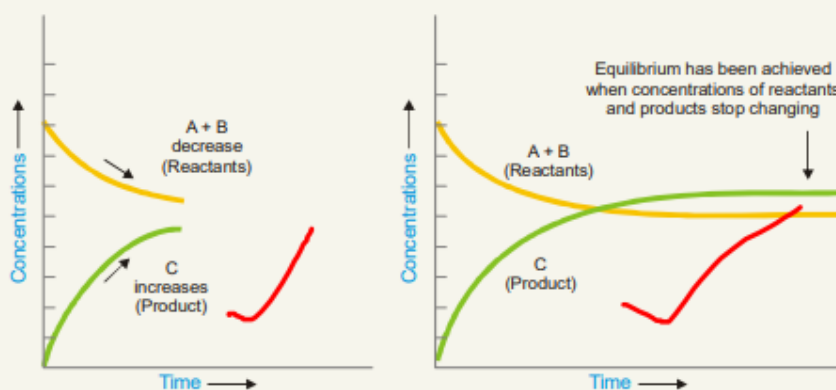
A reaction which can go in the forward and backward direction simultaneously is called a Reversible reaction. Such a reaction is represented by writing a pair of arrows between the reactants and products.



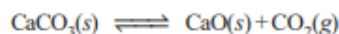
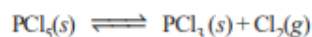
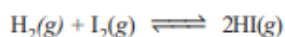
The arrow pointing right indicates the forward reaction, while that pointing left shows the reverse reaction.

Some Examples of Reversible Reactions

A few common examples of reversible reactions are listed below:



A “nuts and bolts” representation of chemical equilibrium for a reversible reaction. Initially (a) there are 100 nuts and 100 bolts in the beakers represented in the figure by 10 of each. The concentration is 100 bolts per beaker for both nuts and bolts. As time passes (b) concentrations of reactants decrease and concentrations of products - assembled units - increase. Equilibrium is established (c) when concentrations stop changing. For this equilibrium, there are 40 nuts per beaker, 40 bolts per beaker, and 60 assembled units per beaker. The rate of the forward reaction (assembling units) is equal to the rate of the reverse reaction (taking the units apart).



NATURE OF CHEMICAL EQUILIBRIUM : ITS DEFINITION

Let us consider the reaction



If we start with A and B in a closed vessel, the forward reaction proceeds to form C and D. The concentrations of A and B decrease and those of C and D increase continuously. As a result the rate of forward reaction also decreases and the rate of the reverse reaction increases. Eventually, the rate of the two opposing reactions equals and the system attains a *state of equilibrium*. Thus **Chemical equilibrium** may be defined as: **the state of a reversible reaction when the two opposing reactions occur at the same rate and the concentrations of reactants and products do not change with time.**

Furthermore, the true equilibrium of a reaction can be attained from both sides. Thus the equilibrium concentrations of the reactants and products are the same whether we start with A and B, or C and D.

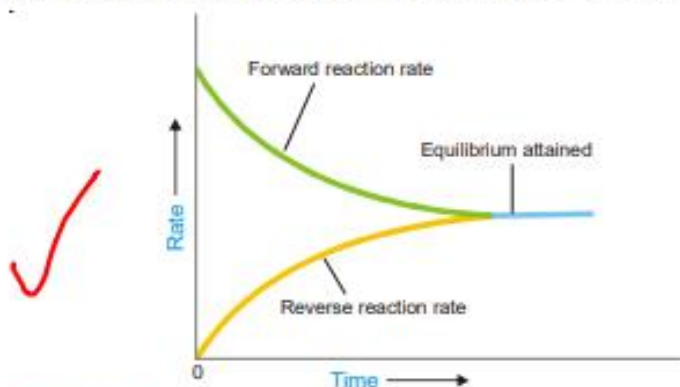


Figure 17.1

At equilibrium the forward reaction rate equals the reverse reaction rate.

Chemical Equilibrium is Dynamic Equilibrium

We have shown above that as the reaction $A + B \rightleftharpoons C + D$ attains equilibrium, the concentrations of A and B, as also of C and D remain constant with time. Apparently it appears that the equilibrium is dead. But it is not so. The equilibrium is dynamic. Actually, the forward and the reverse reactions are taking place at equilibrium but the concentrations remain unchanged.

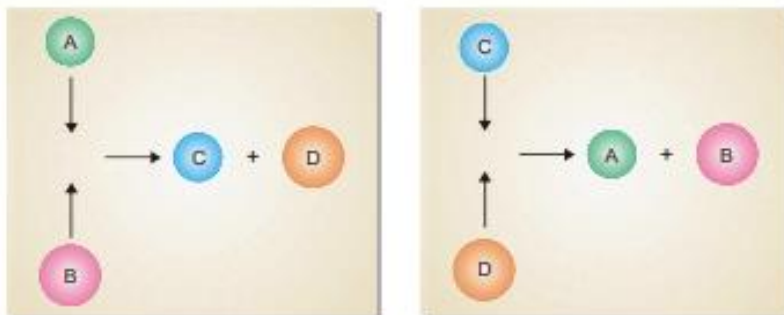


Figure 17.2

Molecules of A and B colliding to give C and D, and those of C and D colliding to give A and B.

(6) At Equilibrium $\Delta G = 0$

At equilibrium the Gibbs free energy (G) is minimum and any change taking place at equilibrium proceeds without change in free energy *i.e.* $\Delta G = 0$.

LAW OF MASS ACTION

Two Norwegian chemists, Guldberg and Waage, studied experimentally a large number of equilibrium reactions. In 1864, they postulated a generalisation called the **Law of Mass action**. It states that : the rate of a chemical reaction is proportional to the active masses of the reactants.

By the term 'active mass' is meant the molar concentration *i.e.*, number of moles per litre. It is expressed by enclosing the formula of the substance in square brackets. Thus a gas mixture containing 1 g of hydrogen (H_2) and 127 g of iodine (I_2) per litre has the concentrations

$$[H_2] = 0.5 \text{ and } [I_2] = 0.5$$

EQUILIBRIUM CONSTANT : EQUILIBRIUM LAW

Let us consider a general reaction



and let $[A]$, $[B]$, $[C]$ and $[D]$ represent the molar concentrations of A, B, C and D at the equilibrium point. According to the Law of Mass action.

$$\text{Rate of forward reaction} \propto [A][B] = k_1 [A][B]$$

$$\text{Rate of reverse reaction} \propto [C][D] = k_2 [C][D]$$

where k_1 and k_2 are rate constants for the forward and reverse reactions.

At equilibrium, rate of forward reaction = rate of reverse reaction.

Therefore,

$$k_1 [A][B] = k_2 [C][D]$$

$$\text{or} \quad \frac{k_1}{k_2} = \frac{[C][D]}{[A][B]} \quad \dots (1)$$

At any specific temperature k_1/k_2 is constant since both k_1 and k_2 are constants. The ratio k_1/k_2 is called **Equilibrium constant** and is represented by the symbol K_c or simply k . The subscript 'c' indicates that the value is in terms of concentrations of reactants and products. The equation (1) may be written as

$$\text{Equilibrium constant} \quad K_c = \frac{[C][D]}{[A][B]}$$

Product concentrations
Reactant concentrations

This equation is known as the **Equilibrium constant expression or Equilibrium law**.

Consider the reaction



Here, the forward reaction is dependent on the collisions of each of two A molecules. Therefore, for writing the equilibrium expression, each molecule is regarded as a separate entity i.e.,



Then the equilibrium constant expression is

$$K_c = \frac{[C][D]}{[A][A]} = \frac{[C][D]}{[A]^2} \quad \text{Power equal to coefficient of A}$$

As a general rule, if there are two or more molecules of the same substance in the chemical equation, its concentration is raised to the power equal to the numerical coefficient of the substance in the equation.

Equilibrium Constant Expression for a Reaction in General Terms

The general reaction may be written as



where a , b , c and d are numerical quotients of the substance, A, B, C and D respectively. The equilibrium constant expression is

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

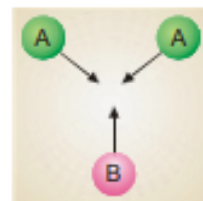


Figure 17.8

SOLVED PROBLEM 3. What is the equilibrium constant expression of the reaction



SOLUTION

Write the product of concentrations of 'products' divided by the product of concentrations of 'reactants'. The concentration of H_2 is to be raised by its quotient in the equation. Thus, the equilibrium constant expression is:

$$K_c = \frac{[\text{CO}][\text{H}_2]^3}{[\text{CH}_4][\text{H}_2\text{O}]}$$

EQUILIBRIUM CONSTANT EXPRESSION IN TERMS OF PARTIAL PRESSURES

When all the reactants and products are gases, we can also formulate the equilibrium constant expression in terms of partial pressure. The relationship between the partial pressure (p) of any one gas in the equilibrium mixture and the molar concentration follows from the general ideal gas equation

$$pV = nRT \quad \text{or} \quad p = \left(\frac{n}{V}\right)RT$$

The quantity $\frac{n}{V}$ is the number of moles of the gas per unit volume and is simply the molar concentration. Thus,

$$p = (\text{Molar concentration}) \times RT$$

i.e., the partial pressure of a gas in the equilibrium mixture is directly proportional to its molar concentration at a given temperature. Therefore, we can write the equilibrium constant expression in terms of partial pressure instead of molar concentrations. For a general reaction

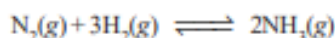


the equilibrium law or the equilibrium constant may be written as

$$K_p = \frac{(p_Y)^y (p_Z)^z}{(p_L)^l (p_M)^m}$$

Here K_p is the equilibrium constant, the subscript p referring to partial pressure. Partial pressures are expressed in atmospheres.

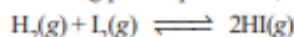
SOLVED PROBLEM 1. Write the equilibrium constant expression for the synthesis of ammonia,



SOLUTION

$$K_p = \frac{(p_{\text{NH}_3})^2}{(p_{\text{N}_2})(p_{\text{H}_2})^3}$$

SOLVED PROBLEM 2. Using partial pressures, write the equilibrium law for the reaction



SOLUTION

$$K_p = \frac{(p_{\text{HI}})^2}{(p_{\text{H}_2})(p_{\text{I}_2})}$$

SOLVED PROBLEM 3. What is the expression K_p for the reaction



SOLUTION

For K_p we use the partial pressure in the equilibrium constant expression. Therefore,

$$K_p = \frac{(p_{\text{NO}_2})^2}{(p_{\text{N}_2\text{O}_4})}$$

HOW K_c AND K_p ARE RELATED?

Let us consider a general reaction



where all reactants and products are gases. We can write the equilibrium constant expression in terms of partial pressures as

$$K_p = \frac{(p_C)^l (p_D)^m}{(p_A)^j (p_B)^k} \quad \text{---(1)}$$

Assuming that all these gases constituting the equilibrium mixture obey the ideal gas equation, the partial pressure (p) of a gas is

$$p = \left(\frac{n}{V} \right) RT$$

Where $\frac{n}{V}$ is the molar concentration. Thus the partial pressures of individual gases, A, B, C and D are:

$$p_A = [\text{A}] RT; p_B = [\text{B}] RT; p_C = [\text{C}] RT; p_D = [\text{D}] RT$$

Substituting these values in equation (1), we have

$$K_p = \frac{[\text{C}]^l (RT)^l [\text{D}]^m (RT)^m}{[\text{A}]^j (RT)^j [\text{B}]^k (RT)^k}$$

or

$$K_p = \frac{[\text{C}]^l [\text{D}]^m}{[\text{A}]^j [\text{B}]^k} \times \frac{(RT)^{l+m}}{(RT)^{j+k}}$$

$$K_p = K_c \times (RT)^{(l+m)-(j+k)}$$

$\therefore$

$$K_p = K_c \times (RT)^{\Delta n} \quad \text{---(2)}$$

where $\Delta n = (l+m) - (j+k)$, the difference in the sums of the coefficients for the gaseous products and reactants.

From the expression (2) it is clear that when $\Delta n = 0$, $K_p = K_c$

SOLVED PROBLEM 1. At 500°C , the reaction between N_2 and H_2 to form ammonia has $K_c = 6.0 \times 10^{-2}$. What is the numerical value of K_p for the reaction?

SOLUTION

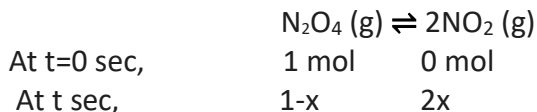
Here, we will use the general expression

$$K_p = K_c (RT)^{\Delta n}$$

For the reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$



→ Let us consider the following reaction takes in a container (V) at a pressure (P) and Temperature (T).



Here, Concentration of N_2O_4 at equilibrium, $[\text{N}_2\text{O}_4] = \frac{1-x}{V}$

Concentration of NO_2 at equilibrium, $[\text{NO}_2] = \frac{2x}{V}$

Now,

$$\begin{aligned}
 K_c &= \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} \\
 &= \frac{\left(\frac{2x}{V}\right)^2}{\frac{1-x}{V}} \\
 &= \frac{4x^2}{V^2 \cdot \frac{(1-x)}{V}} \\
 &= \frac{4x^2}{(1-x) \cdot V}
 \end{aligned}$$

Unit of K_c = $(\text{mol/L})^{\Delta n} = (\text{mol/L})^{2-1} = (\text{mol/L})^1 = \text{mol/L}$

Again, Total mole, $n = 1-x+2x = 1+x$ mol

Partial Pressure of NO_2 , $P_{\text{NO}_2} = \frac{2x}{1+x} \cdot P$

Partial Pressure of N_2O_4 , $P_{\text{N}_2\text{O}_4} = \frac{1-x}{1+x} \cdot P$

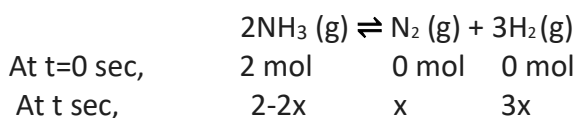
$$\begin{aligned}
 K_p &= \frac{P_{\text{NO}_2}^2}{P_{\text{N}_2\text{O}_4}} \\
 &= \frac{\left(\frac{2x}{1+x} \cdot P\right)^2}{\frac{1-x}{1+x} \cdot P} \\
 &= \frac{4x^2 \cdot P^2}{(1+x)^2 \cdot \left(\frac{1-x}{1+x}\right) \cdot P} \\
 &= \frac{4x^2 \cdot P}{1-x^2}
 \end{aligned}$$

$$\text{Unit of } K_p = (\text{atm})^{\Delta n} = (\text{atm})^{2-1} = (\text{atm})^1 = \text{atm}$$

So, $K_p \neq K_c$; as $\Delta n = 2-1 = 1$



→ Let us consider the following reaction takes in a container (V) at a pressure (P) and Temperature (T).



Here, Concentration of N_2 at equilibrium, $[\text{N}_2] = \frac{x}{V}$

Concentration of H_2 at equilibrium, $[\text{H}_2] = \frac{3x}{V}$

Concentration of NH_3 at equilibrium, $[\text{NH}_3] = \frac{2-2x}{V}$

Now,

$$\begin{aligned}
 K_c &= \frac{[\text{N}_2] [\text{H}_2]^3}{[\text{NH}_3]^2} \\
 &= \frac{\frac{x}{V} \cdot \left(\frac{3x}{V}\right)^3}{\left(\frac{2-2x}{V}\right)^2} \\
 &= \frac{27x^4}{V^4 \cdot \frac{(2-2x)^2}{V^2}} \\
 &= \frac{27x^4}{V^2 \cdot (2-2x)^2}
 \end{aligned}$$

$$\text{Unit of } K_c = (\text{mol/L})^{\Delta n} = (\text{mol/L})^{(1+3)-2} = (\text{mol/L})^2 = \text{mol}^2/\text{L}^2$$

Again, Total mole, $n = 2-2x+x+3x = 2+2x$ mol

$$\text{Partial Pressure of } \text{NH}_3, P_{\text{NH}_3} = \frac{2-2x}{2+2x} \cdot P$$

$$\text{Partial Pressure of } \text{H}_2, P_{\text{H}_2} = \frac{3x}{2+2x} \cdot P$$

$$\text{Partial Pressure of } \text{N}_2, P_{\text{N}_2} = \frac{x}{2+2x} \cdot P$$

$$\begin{aligned}
 K_p &= \frac{P_{H_2}^3 \cdot P_{N_2}}{P_{NH_3}^2} \\
 &= \frac{\left(\frac{3x}{2+2x} \cdot P\right)^3 \cdot \left(\frac{x}{2+2x} \cdot P\right)}{\left(\frac{2-2x}{2+2x} \cdot P\right)^2} \\
 &= \frac{27x^4 \cdot P^4}{(2+2x)^4 \cdot \frac{(2-2x)^2}{(2+2x)^2} P^2} \\
 &= \frac{27x^4 \cdot P^4}{(2+2x)^2 \cdot P^2 \cdot (2-2x)^2} \\
 &= \frac{27x^4 \cdot P^2}{(4-4x^2)^2}
 \end{aligned}$$

Unit of K_p = $(atm)^{\Delta n} = (atm)^{(1+3)-2} = (atm)^2$

So, $K_p \neq K_c$; as $\Delta n = (1+3) - 2 = 2$

******** **Estimate K_p and K_c of the reaction $2K \rightleftharpoons L + 3M$; [$K = NH_3$, $L = N_2$ and $M = H_2$], where the extent of the products at equilibrium is ' x ' mole and the reaction is carried out in a vessel of volume **V** litter at temperature **T** and total pressure **P**.**

we have $\Delta n = (\text{sum of quotients of products}) - (\text{sum of quotients of reactants})$
 $= 2 - 4 = -2$

$$K_c = 6.0 \times 10^{-2}$$

$$T = 500 + 273 = 773 \text{ K}$$

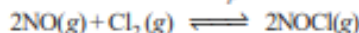
$$R = 0.0821$$

Substituting the value of R , T , K_c and Δn in the general expression, we have

$$K_p = (6.0 \times 10^{-2}) [(0.0821) \times (773)]^{-2}$$

$$= 1.5 \times 10^{-5}$$

SOLVED PROBLEM 2. The value of K_p at 25°C for the reaction



is $1.9 \times 10^3 \text{ atm}^{-1}$. Calculate the value of K_c at the same temperature.

SOLUTION

We can write the general expression as

$$K_p = K_c (RT)^{\Delta n} \text{ or } K_c = \frac{K_p}{(RT)^{\Delta n}}$$

Here,

$$T = 25 + 273 = 298 \text{ K}$$

$$R = 0.0821$$

$$\Delta n = 2 - (2 + 1) = -1$$

$$K_p = 1.9 \times 10^3$$

Substituting these values in the general expression

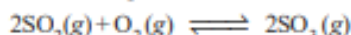
$$K_c = \frac{1.9 \times 10^3}{(0.0821 \times 298)^{-1}}$$

$$= 4.6 \times 10^4$$

Calculation of K_c from Experimental Information

To determine the value of K_c of a reaction, write the balanced equation. Then write the equilibrium constant expression. Substitute in it the equilibrium concentrations of the reactants and products found experimentally. Thus calculate the value of K_c .

SOLVED PROBLEM 1. At equilibrium for the reaction



the concentrations of reactants and products at 727°C were found to be : $\text{SO}_2 = 0.27 \text{ mol L}^{-1}$; $\text{O}_2 = 0.40 \text{ mol L}^{-1}$; and $\text{SO}_3 = 0.33 \text{ mol L}^{-1}$. What is the value of the equilibrium constant, K_c at this temperature?

SOLUTION

Write the equilibrium constant expression for the reaction

$$K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 [\text{O}_2]}$$

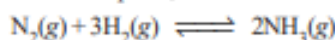
We know that:

$$[\text{SO}_3] = 0.33 \text{ mol L}^{-1}; [\text{SO}_2] = 0.27 \text{ mol L}^{-1}; [\text{O}_2] = 0.40 \text{ mol L}^{-1}$$

Substituting these values in the equilibrium constant expression, we have

$$K_c = \frac{(0.33)^2}{(0.27)^2 \times (0.40)} = 3.7 \text{ mol l}^{-1} \text{ or simply } 3.7$$

SOLVED PROBLEM 2. Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C . When equilibrium is established, 3.00 moles of N_2 , 2.10 moles of H_2 and 0.298 mole of NH_3 are found to be present. Find the value of K_c for the reaction



at 500°C .

SOLUTION

The equilibrium concentrations are obtained by dividing the number of moles of each reactant and product by the volume, 5.00 litres. Thus,

$$[\text{N}_2] = 3.00 \text{ mole}/5.00 \text{ L} = 0.600 \text{ M}$$

$$[\text{H}_2] = 2.10 \text{ mole}/5.00 \text{ L} = 0.420 \text{ M}$$

$$[\text{NH}_3] = 0.298 \text{ mole}/5.00 \text{ L} = 0.0596 \text{ M}$$

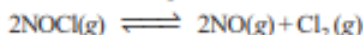
Substituting these concentrations (not number of moles) in the equilibrium constant expression, we get the value of K_c .

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \frac{(0.0596)^2}{(0.600)(0.420)^3} = 0.080$$

Thus, for the reaction of H_2 and N_2 to form NH_3 at 500°C , we can write

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = 0.080$$

SOLVED PROBLEM 3. Calculate the equilibrium constant at 25°C for the reaction



In an experiment, 2.00 moles of NOCl were placed in a one-litre flask and the concentration of NO after equilibrium achieved was 0.66 mole/litre.

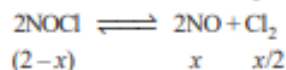
SOLUTION

Let us write the equilibrium constant expression for the balanced chemical equation.

$$K_c = \frac{[\text{NO}]^2 [\text{Cl}_2]}{[\text{NOCl}]^2}$$

Next we proceed to find the equilibrium concentrations of NOCl , NO and Cl_2 . Since the volume of the reaction vessel is one litre, the moles of various reagents also represent their molar concentrations.

The initial amount of NOCl is 2.00 moles. Let us assume that x moles of it have reacted to achieve the equilibrium. The moles of different reagents at equilibrium are indicated in the equation



According to this equation, one mole of NOCl gives one mole of NO and half a mole of Cl_2 .

The experimental value of equilibrium concentration of NO is 0.66 mole/litre. Therefore, the

equilibrium concentration Cl_2 is half of it i.e. 0.33 mole/litre. Since $x = 0.66$, the equilibrium concentration of $\text{NOCl} = (2 - 0.66) = 1.34$. To find K_c , these values of equilibrium concentrations are substituted in the equilibrium constant expression.

$$K_c = \frac{[\text{NO}]^2 [\text{Cl}_2]}{[\text{NOCl}]^2} = \frac{(0.66)^2 (0.33)}{(1.34)^2} = 0.080$$

SOLVED PROBLEM 4. At a certain temperature, 0.100 mole of H_2 and 0.100 mole of I_2 were placed in a one-litre flask. The purple colour of iodine vapour was used to monitor the reaction. After a time, the equilibrium



was established and it was found that the concentration of I_2 decreased to 0.020 mole/litre. Calculate the value of K_c for the reaction at the given temperature.

SOLUTION

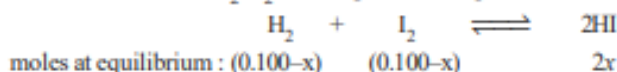
The equilibrium constant expression for the reaction



is
$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

Let us proceed to find the equilibrium concentrations of the various species.

The initial moles of H_2 , I_2 and HI present at equilibrium are



Since the volume of the reaction vessel is one litre, the moles at equilibrium also represent the equilibrium concentrations. Thus,

$$[\text{I}_2] = 0.100 - x = 0.020 \text{ (given)}$$

or
$$\begin{aligned} x &= 0.100 - 0.020 \\ &= 0.080 \end{aligned}$$

Knowing the value of x , we can give the equilibrium concentrations of H_2 , I_2 and HI .

$$[\text{H}_2] = 0.100 - x = 0.100 - 0.080 = 0.020$$

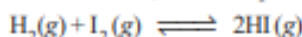
$$[\text{I}_2] = 0.020 \text{ (given)}$$

$$[\text{HI}] = 2x = 0.160$$

Substituting these values in the equilibrium constant expression, we get the value of K_c .

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(0.160)^2}{0.02 \times 0.02} = 64$$

SOLVED PROBLEM 5. 13.5 ml of HI are produced by the interaction of 8.1 ml of hydrogen and 9.3 ml of iodine vapour at 444°C . Calculate the equilibrium constant at this temperature of the reaction.



SOLUTION

The equilibrium constant expression for the reaction is

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

Let us proceed to find the equilibrium concentrations.

SOLVED PROBLEM 2. At a certain temperature, K for the reaction $3\text{C}_2\text{H}_2(\text{g}) \rightleftharpoons \text{C}_6\text{H}_6(\text{g})$ is 4. If the equilibrium concentration of C_2H_2 is 0.5 mole/litre, what is the concentration of C_6H_6 ?

SOLUTION

The reaction can be written as ,



The equilibrium constant expression is

$$K = \frac{[\text{C}_6\text{H}_6]}{[\text{C}_2\text{H}_2]^3}$$

It is given that:

$$K = 4$$

$$[\text{C}_2\text{H}_2] = 0.5 \text{ mole/litre}$$

Substituting these values,

$$4 = \frac{[\text{C}_6\text{H}_6]}{[0.5]^3}$$

$$\therefore [\text{C}_6\text{H}_6] = 4 \times (0.5)^3 = \mathbf{0.5 \text{ moles/litre}}$$

SOLVED PROBLEM 3. For the reaction



the value of K at 552°C is 0.137. If 5 moles of CO_2 , 5.0 moles of H_2 , 1.0 mole of CO and 1.0 mole of H_2O are initially present, what are the actual concentrations of CO_2 , H_2 , CO and H_2O at equilibrium?

SOLUTION

If x moles of CO_2 has reacted at equilibrium, x mole of CO and x mole of H_2O will be produced. The moles at equilibrium due to the reaction will be



$$\text{Moles at equilibrium : } \begin{array}{cccc} 5-x & 5-x & x & x \end{array}$$

But 1 mole each of CO and H_2O were present initially. Therefore the actual moles at equilibrium would be

$$\text{CO}_2 = 5 - x$$

$$\text{H}_2 = 5 - x$$

$$\text{CO} = 1 + x$$

$$\text{H}_2\text{O} = 1 + x$$

If V litres be the volume of the reaction vessel, the concentration of the various species are:

$$[\text{CO}_2] = (5 - x)/V$$

$$[\text{H}_2] = (5 - x)/V$$

$$[\text{CO}] = (1 + x)/V$$

$$[\text{H}_2\text{O}] = (1 + x)/V$$

Substituting the values in the equilibrium constant expression, we have

$$K = \frac{[\text{CO}][\text{H}_2\text{O}]}{[\text{CO}_2][\text{H}_2]}$$

$$0.137 = \frac{\left[\frac{(1+x)}{V} \right] \left[\frac{(1+x)}{V} \right]}{\left[\frac{(5-x)}{V} \right] \left[\frac{(5-x)}{V} \right]} = \frac{(1+x)^2 V^2}{(5-x)^2 V^2}$$

$$0.137 = \frac{(1+x)^2}{(5-x)^2} = \frac{1+2x+x^2}{25-10x+x^2}$$

or
$$0.137(25) - 0.137(10)x + 0.137x^2 = 1 + 2x + x^2$$

$$0.863x^2 + 3.370x - 2.425 = 0$$

Solving for x using the quadratic formula

$$x = 0.62$$

Thus the final equilibrium concentrations are:

$$[\text{CO}] = 1 + x = 1 + 0.62 = 1.62 \text{ mole l}^{-1}$$

$$[\text{H}_2\text{O}] = 1 + x = 1 + 0.62 = 1.62 \text{ mole l}^{-1}$$

$$[\text{CO}_2] = 5 - x = 5 - 0.62 = 4.38 \text{ mole l}^{-1}$$

$$[\text{H}_2] = 5 - x = 5 - 0.62 = \mathbf{4.38 \text{ mole l}^{-1}}$$

SOLVED PROBLEM 4. For the reaction $\text{I}_2(\text{g}) \rightleftharpoons 2\text{I}(\text{g})$, $K = 3.76 \times 10^{-5}$ at 727°C . Let 1 mole of I_2 be injected into a 2-litre glass box at 727°C . What will be the concentrations of I_2 and I when the equilibrium is attained?

SOLUTION

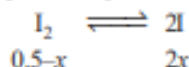
The balanced equation for the reaction is



1 mole of I_2 is injected in a 2-litre box. Thus the initial concentration of I_2 is

$$\frac{1 \text{ mole}}{2 \text{ litre}} = 0.5 \text{ M}$$

Let x moles of I_2 decompose at equilibrium. Then the moles of the two species are:



Therefore the equilibrium concentrations are:

$$[\text{I}_2] = 0.5 - x$$

$$[\text{I}] = 2x$$

and

$$K = 3.76 \times 10^{-5}$$

Substituting the values in the equilibrium constant expression, we have

$$K = \frac{[\text{I}]^2}{[\text{I}_2]} = 3.76 \times 10^{-5}$$

or
$$\frac{(2x)^2}{(0.5-x)} = 3.76 \times 10^{-5}$$

Solving for x , using the quadratic equation formula,

$$x = 2.17 \times 10^{-3}$$

Thus the concentrations of I and I_2 at equilibrium are:

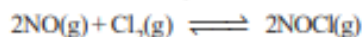
$$[\text{I}] = 2x = \mathbf{4.34 \times 10^{-3} \text{ mole l}^{-1}}$$

$$[\text{I}_2] = 0.5 - x = \mathbf{0.498 \text{ mole l}^{-1}}$$

CALCULATIONS INVOLVING K_p

It may be noted that the value of K_p depends on the units used to express pressures, which are usually atmospheres.

SOLVED PROBLEM 1. The reaction



was studied at 25°C. The partial pressures at equilibrium were found to be

$$p_{\text{NOCl}} = 1.2 \text{ atm}$$

$$p_{\text{NO}} = 5.0 \times 10^{-2} \text{ atm}$$

$$p_{\text{Cl}_2} = 3.0 \times 10^{-1} \text{ atm}$$

Calculate K_p for the reaction at 25°C.

SOLUTION

The equilibrium constant expression in terms of partial pressures is

$$K_p = \frac{(p_{\text{NOCl}})^2}{(p_{\text{NO}})^2 (p_{\text{Cl}_2})}$$

Substituting the values of partial pressures

$$K_p = \frac{(1.2)^2}{(5.0 \times 10^{-2})^2 \times (3.0 \times 10^{-1})} = 1.92 \times 10^3$$

SOLVED PROBLEM 2. At 60°C and a total pressure of 1 atmosphere dinitrogen tetroxide, N_2O_4 , is 50% dissociated into nitrogen dioxide, NO_2 .



Calculate the value of K_p at this temperature.

SOLUTION

Let us have 1 mole of N_2O_4 to start with of which x mole dissociates at equilibrium. Then the moles of N_2O_4 and NO_2 at equilibrium are:



moles at equilibrium : $(1-x)$ $2x$

Since N_2O_4 is 50% dissociated, $x = 0.5$ mole and the equilibrium mixture contains

$$\text{NO}_2 = 2x = 2 \times 0.5 = 1 \text{ mole}$$

$$\text{N}_2\text{O}_4 = (1-x) = (1-0.5) = 0.5 \text{ mole}$$

That is, the moles of N_2O_4 and NO_2 are present in the ratio 1 : 2

$\therefore$ The partial pressure of $\text{N}_2\text{O}_4 = \frac{1}{3} \text{ atm}$

The partial pressure of $\text{NO}_2 = \frac{2}{3} \text{ atm}$

Substituting these values in the equilibrium constant expression, we have

$$K_p = \frac{(p_{\text{NO}_2})^2}{p_{\text{N}_2\text{O}_4}} = \frac{(2/3)^2}{1/3} = \frac{4}{3} = 1.33$$

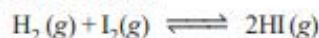
✓ UNITS OF EQUILIBRIUM CONSTANT

In the equilibrium expression for a particular reaction, the concentrations are given in units of moles/litre or mol/L, and the partial pressure are given in atmospheres (atm). The units of K_c and K_p depend on the specific reaction.

✓ (1) When the total number of moles of reactants and products are equal

In the equilibrium expression of these reactions, the concentration or pressure terms in the numerator and denominator exactly cancel out. Thus K_c or K_p for such a reaction is without units.

Taking example of the reaction .



$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(\text{mol/L})^2}{(\text{mol/L})(\text{mol/L})} \quad (\text{No units})$$

$$K_p = \frac{(P_{\text{HI}})^2}{(P_{\text{H}_2})(P_{\text{I}_2})} = \frac{(\text{atm})^2}{(\text{atm})(\text{atm})} \quad (\text{No units})$$

✓ (2) When the total number of moles of the reactants and products are unequal

In such reactions K_c will have units $(\text{mol/litre})^n$ and K_p will have units $(\text{atm})^n$ where n is equal to the total number of moles of products minus the total number of moles of reactants. Thus for the reaction



$$K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = \frac{(\text{mol/L})^2}{(\text{mol/L})} = \text{mol/L}$$

$$K_p = \frac{(p_{\text{NO}_2})^2}{(P_{\text{N}_2\text{O}_4})} = \frac{(\text{atm})^2}{(\text{atm})} = \text{atm}$$

Thus units for K_c are mol/L and for K_p units are atm. For the reaction



the units for K_c and K_p may be found as follows

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \frac{(\text{mol/L})^2}{(\text{mol/L})(\text{mol/L})^3} = \text{mol}^{-2} \text{L}^2$$

$$K_p = \frac{(p_{\text{NH}_3})^2}{(p_{\text{N}_2})(p_{\text{H}_2})^3} = \frac{(\text{atm})^2}{(\text{atm})(\text{atm})^3} = \text{atm}^{-2}$$

Thus units of K_c are $\text{mol}^{-2} \text{L}^2$ and units of K_p are atm^{-2} .

✓ It may be noted, however, that the units are often omitted when equilibrium constants are listed in tables.

✓ Thermodynamic Derivation of Law of Chemical Equilibrium

Let us consider a general reaction



The chemical potential of a substance in a mixture is related to its activity by the expression

$$\mu = \mu^\circ + RT \ln a \quad \dots (i)$$

where μ° is the chemical potential of the pure substance in standard state of unit activity, R is gas constant and T the absolute temperature.

For a mole of the substance A we can write using the equation (i)

$$a\mu_A = a(\mu^\circ + RT \ln a_A)$$

and similarly

$$b\mu_B = b(\mu^\circ + RT \ln a_B)$$

$$c\mu_C = c(\mu^\circ + RT \ln a_C)$$

$$d\mu_D = d(\mu^\circ + RT \ln a_D)$$

The change in free energy for the reaction is given by

$$\Delta G = G_{\text{products}} - G_{\text{reactants}}$$

On substitution we get

$$\begin{aligned} \Delta G &= (c\mu_C + d\mu_D + \dots) - (a\mu_A + b\mu_B + \dots) \\ &= [c\{\mu_C^\circ + RT \ln a_C\} + d\{\mu_D^\circ + RT \ln a_D\}] \\ &\quad - [a\{\mu_A^\circ + RT \ln a_A\} + b\{\mu_B^\circ + RT \ln a_B\}] \\ &= [\{c\mu_C^\circ + d\mu_D^\circ + \dots\} - \{a\mu_A^\circ + b\mu_B^\circ + \dots\}] + RT \ln \frac{a_C^c \times a_D^d \times \dots}{a_A^a \times a_B^b \times \dots} \\ \Delta G &= \Delta G^\circ + RT \ln \frac{a_C^c \times a_D^d \times \dots}{a_A^a \times a_B^b \times \dots} \quad \dots (ii) \end{aligned}$$

where ΔG° is the difference in free energy of the reaction when all reactants and products are in their standard state. It is given by

$$\Delta G^\circ = \{c\mu_C^\circ + d\mu_D^\circ + \dots\} - \{a\mu_A^\circ + b\mu_B^\circ + \dots\}$$

In equation (ii) the factor A given by

$$\frac{a_C^c \times a_D^d \times \dots}{a_A^a \times a_B^b \times \dots}$$

stands for the reaction quotient of activities of the product and reactants. It may be denoted by J .

The equation (ii) becomes

$$\Delta G = \Delta G^\circ + RT \ln J \quad \dots (iii)$$

The equation (iii) is called van't Hoff reaction isotherm

For the reaction at equilibrium

$$\Delta G = 0$$

therefore,

$$\Delta G^\circ = -RT \ln J$$

or

$$\Delta G = -RT \ln \frac{a_C^c \times a_D^d \times \dots}{a_A^a \times a_B^b \times \dots}$$

As ΔG° is the free energy of the reaction in the standard state and is constant at a given temperature.

Also, the gas constant R and T are constant, the factor

$$\frac{a_C^c \times a_D^d \times \dots}{a_A^a \times a_B^b \times \dots}$$

is a constant i.e.

$$\frac{a_C^c \times a_D^d \times \dots}{a_A^a \times a_B^b \times \dots} = \text{a constant} = K$$

It is nothing but the law of chemical equilibrium. Therefore, from equation (iii) we have

$$\Delta G^\circ = -RT \ln K \quad \dots (iv)$$

or

$$\Delta G^\circ = -2.303 RT \log K \quad \dots (v)$$

The equation (iv) is also called van't Hoff Isotherm. It may be used to calculate the change in free energy of a reaction in the standard (ΔG°) from the equilibrium constant and vice-versa.

The sign of ΔG° indicates whether the forward or reverse reaction is spontaneous. Considering the equation (iv), we can have three possibilities depending on the sign of ΔG° for the reaction.

(1) If ΔG° is **negative**, $\log K$ must be positive and the reaction proceeds spontaneously in the forward reaction.

(2) If ΔG° is **positive**, $\log K$ must be negative and K is less than one. The reverse reaction is then spontaneous.

(3) If $\Delta G^\circ = 0$, $\log K = 0$ and $K = 1$. The reaction is at equilibrium.

Meaning of the Magnitude of the Free Energy Change

If ΔG° is a large negative number, K will be much greater than one and the forward reaction will proceed nearly to completion. On the other hand if ΔG° is a large positive number, K will be a small fraction. Then the reverse reaction will proceed almost to completion.

SOLVED PROBLEM 1. Calculate K for reaction which has ΔG° value -20 kcal at 25°C .

SOLUTION

We know that

$$\Delta G^\circ = -2.303 RT \log K \quad \dots (a)$$

where ΔG° is standard free energy; R is gas constant; T is absolute temperature; and K is equilibrium constant. It must be remembered that the same units of ΔG° , R and T are to be used in a given problem.

If ΔG° is given in calories

$$R = 1.99$$

$$T = 25 + 273 = 298 \text{ K}$$

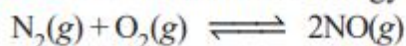
The value of K from expression (a) may be calculated as

$$\begin{aligned} -\log K &= \frac{-\Delta G^\circ}{(2.303)(1.99)(298)} \\ &= \frac{(20,000)}{1365.75} = 14.7 \end{aligned}$$

Taking antilogs, $\log K = 0.7 + 14.0$

Thus $K = 5 \times 10^{14}$

SOLVED PROBLEM 2. The standard free energy change for the reaction



is $+173.1$ kJ. Calculate K_p for the reaction at 25°C .

SOLUTION

Here, we will use the expression

$$\Delta G^\circ = -2.303 RT \log K_p$$

or

$$\log K_p = -\frac{\Delta G^\circ}{2.303 RT}$$

Substituting the values

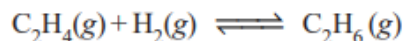
$$\log K_p = \frac{1.73 \times 10^5}{(2.303)(8.314)(298)}$$

$$= -30.34$$

Taking antilogs

$$\begin{aligned} K_p &= 10^{0.66} \times 10^{-31} \\ &= \mathbf{4.6 \times 10^{-31}} \end{aligned}$$

SOLVED PROBLEM 3. The equilibrium constant, K_p for the reaction



is $5.04 \times 10^{17} \text{ atm}^{-1}$ at 25°C . Calculate ΔG° .

SOLUTION

$$\Delta G^\circ = -2.303 RT \log K_p \quad \dots(a)$$

Here, $R = 8.314$; $T = 25 + 273 = 298 \text{ K}$; $K_p = 5.04 \times 10^{17} \text{ atm}^{-1}$

Substituting the values in the expression (a)

$$\begin{aligned} \Delta G^\circ &= -2.303 (8.314) (298) \log (5.04 \times 10^{17}) \\ &= -5706 \times \log (5.04 \times 10^{17}) \\ &= -5706 \times (17.702) \\ &= -1.010 \times 10^5 \text{ J} \end{aligned}$$

Thus

$$\Delta G^\circ = - \mathbf{101 \text{ kJ}}$$